

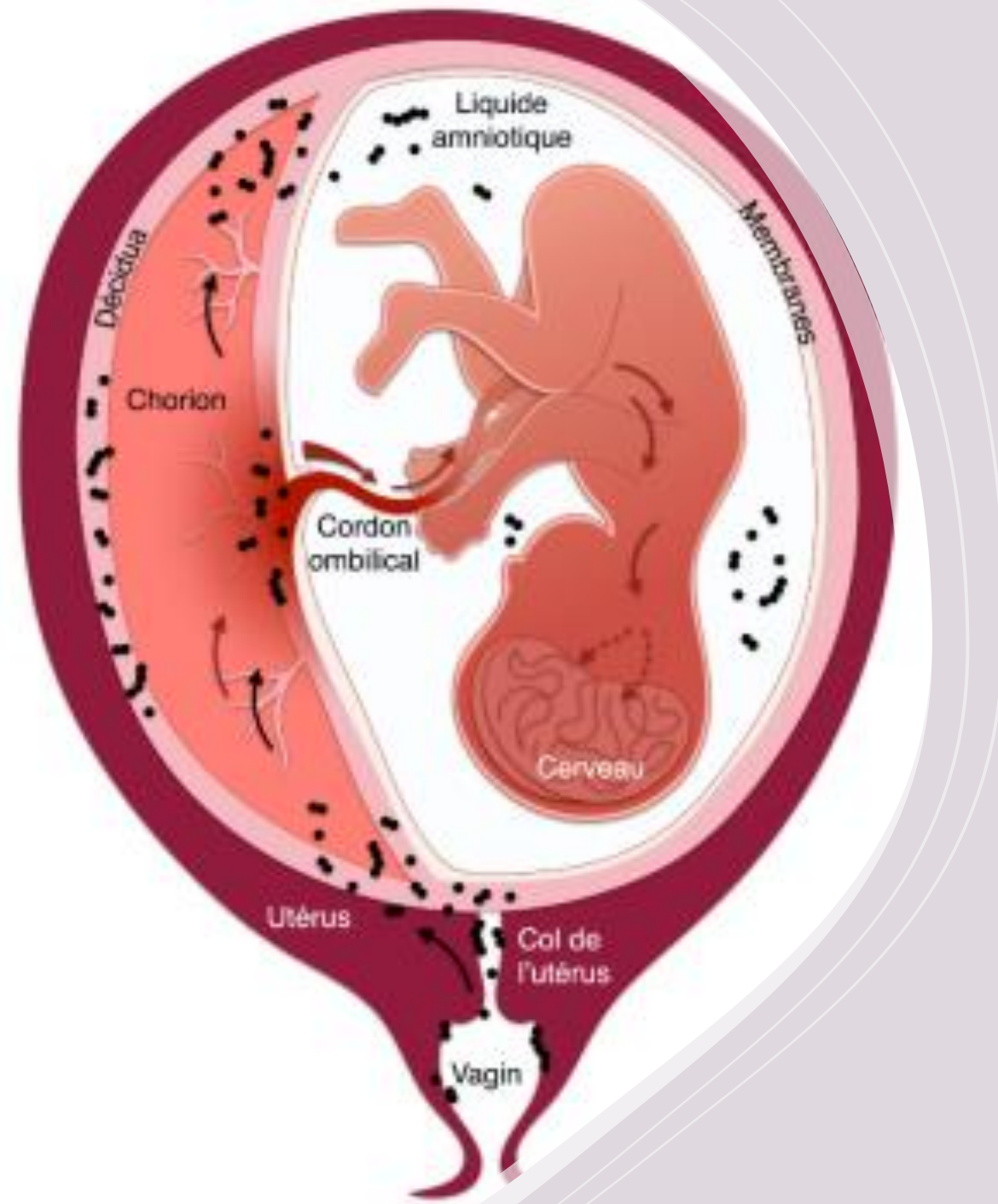
PED-ID

Interesting Case Conference

Pattranan Kusontammarat, MD

Division of Pediatric Infectious Diseases,
Department of Pediatrics

Faculty of Medicine, Chiang Mai University



Identification Data

- **HN** 4237133
- **Case** preterm female newborn GA 29+2 wk
- **Date of birth:** 7/6/67 Time : 4.47 pm
- **Place of birth:** Maharaj Nakhon Chiang Mai
- **Birth bodyweight:** 1,360 gm (AGA)

Maternal History

- **Maternal age:** 36 year-old. G2P0101 GA 29+2 week
- **G1** Last 12 years; preterm 35 weeks BBW 1,900 gm
- **G2** New family Late ANC: 1st ANC at GA 28+6 weeks at รพ แม่และเด็ก, Total ANC 1 ครั้ง
- **Serology:** Anti-HIV NR/NR, HBs Ag neg, VDRL NR/NR
- Blood group O, Rh pos , Hb 9.6/10 , Hct 30/29.7% , MCV 90, not couple at risk for severe thalassemia

Maternal History

Prenatal risk:

PROM 12 hr (Did not receive antibiotic prophylaxis)

Maternal suspected triple I

- Fever + CBC: leukocytosis WBC 20,680 cells/cu.mm. (Neutrophil 87.8%, Lymphocyte 5.3%
platelet 240,000 cells/cu.mm) + Fetal tachycardia (FHR 150-160 bpm)

Problem Lists

Preterm female newborn GA 29+2 wk, NL, BBW 1,360 gm (AGA) Apgar 7-8-8T

Respiratory distress due to RDS S/P curosurf at age 2 hrs

Severe persistent pulmonary hypertension with PDA

GBS sepsis with septic shock with DIC

Neonatal jaundice suspected due to sepsis S/P phototherapy 9-18/6/67

Post Natal History

At birth

RDS with respiratory failure

Severe PPHN

Symptomatic infant at risk EONS
(spontaneous preterm labor, maternal
suspected triple I)

Management: surfactant at age 2 hr

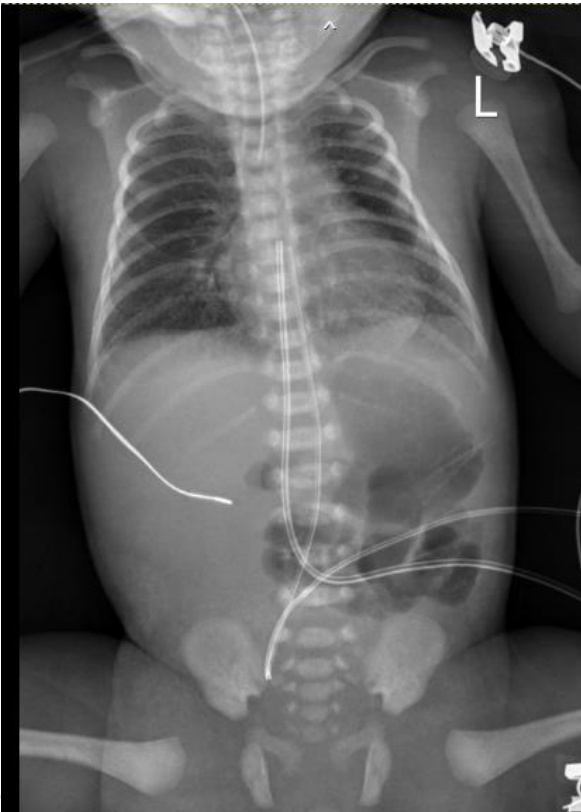
On ventilator support

Empirical antibiotic:
ampicillin, gentamycin



Timeline

8/6/67
DOL 1



**# GBS septicemia with DIC
with pulmonary hemorrhage**

Mx: on HFOV

Inotrope: Levophred+Primacor

Ampicillin(300MKDay) q 8 hr

Cefotaxime (150 MKDay) q 8 hr

H/C (UAC , peripheral):

#	Qty	Code	Organism
1		SCCGB	Streptococcus agalactiae

Time to positive 14 hours

#	Code	Antibiotic	MIC	Result
1	P	Penicillin	≤ 0.06	S
2	TX	Ceftriaxone	≤ 0.12	S
3	LX	Levofloxacin	1	S
4	E	Erythromycin	≤ 0.12	S
5	CC	Clindamicin	≤ 0.25	S
6	VA	Vancomycin	0.5	S

Lab investigations

CBC	7/6/67	8/6/67
Hb (g/dl)	13.9	14.1
Hct (%)	40.3	42.3
WBC (cell/cu.mm.)	2,960	5,150
Neutrophil (%)	33.1	62.3
Lymphocyte(%)	60.5	30.4
Monocyte(%)	5.1	5.2
ANC	900	3210
Platelet(cell/cu.mm.)	150,000	117,000
MCV	100	108

Chemistry	8/6/67
BUN	29
Creatinine	0.95
Na	123
K	4.7
Cl	94
CO2	13
Ca	6.6
Mg	1.77
P	6.5

Sepsis screen	12 hr	24 hr
ANC	900 (3,500-21,000)	3,210 (3,000-22,000)
Hs-CRP (mg/L)	4.74	30.6
I:T ratio	0.06	0.28
mESR	3 mm	-

Timeline

DOB
7/6/67

8/6/67
DOL 1

13/6/67
DOL 6

- # RDS # Severe PPHN
- # GBS septicemia with DIC with pulmonary hemorrhage
- **Hemoculture:** GBS

- # **GBS septicemia with DIC with pulmonary hemorrhage**
- **Hemoculture:** NG

- # VAP
- New onset fever
- +CXR new infiltration
- **Urine G/S:** no bacteria
- **Urine C/S:** NG
- **Sputum G/S:** no bacteria
- **Sputum C/S:** NG
- **Hemoculture:** NG

Ampicillin(50MKDose) q 12 hr

Ampicillin(300MKDay) q 8 hr

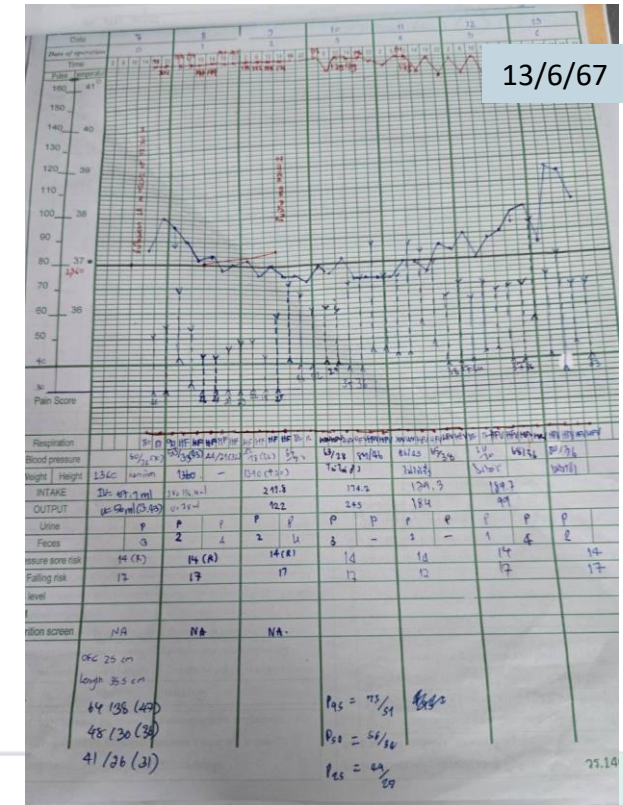
Gentamycin 5 MKDose

Cefotaxime (150 MKDay) q 8 hr

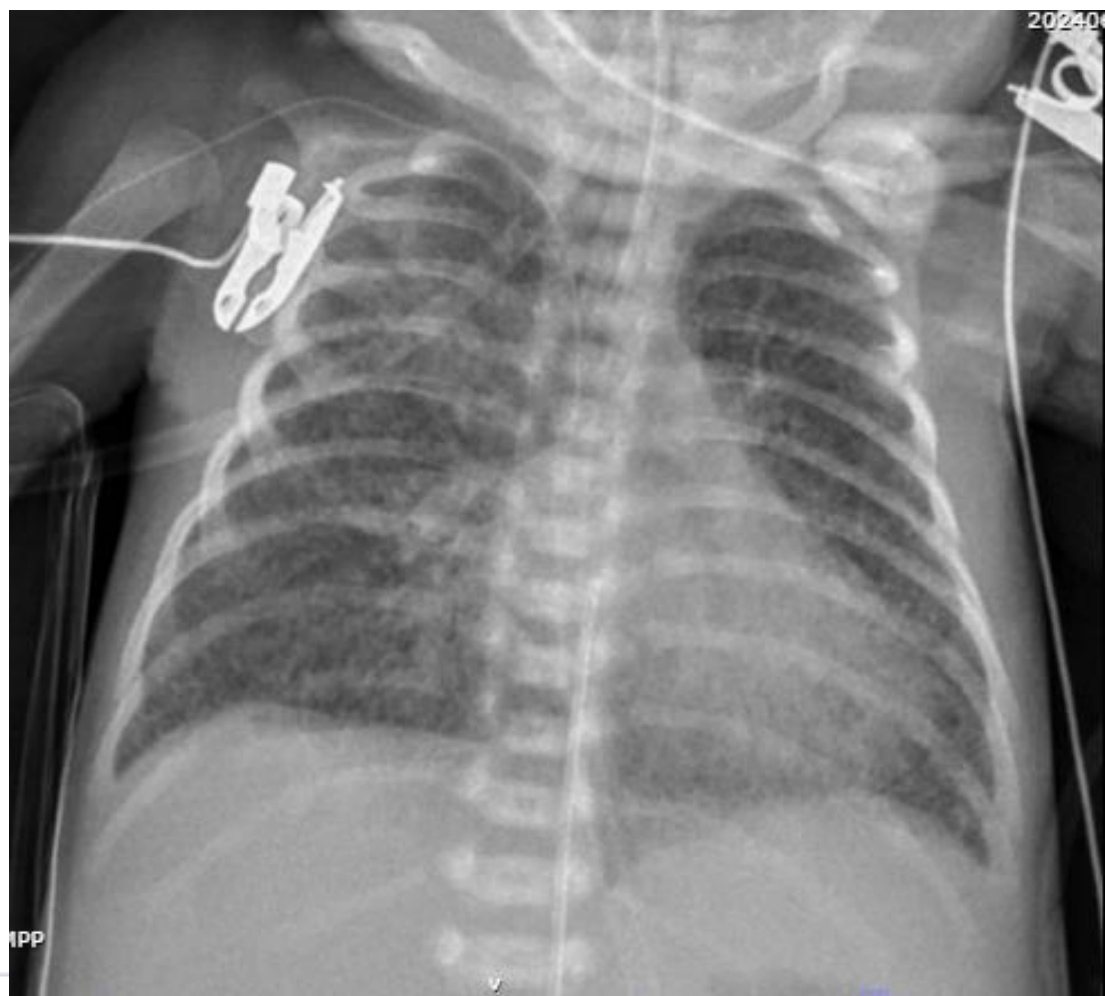
Meropenem(40MKDose) q 12 hr

Vancomycin 72 hr

Line UVC (7-12/6/67)
UAC (7-12/6/67)
PICC line (12/6/67-now)



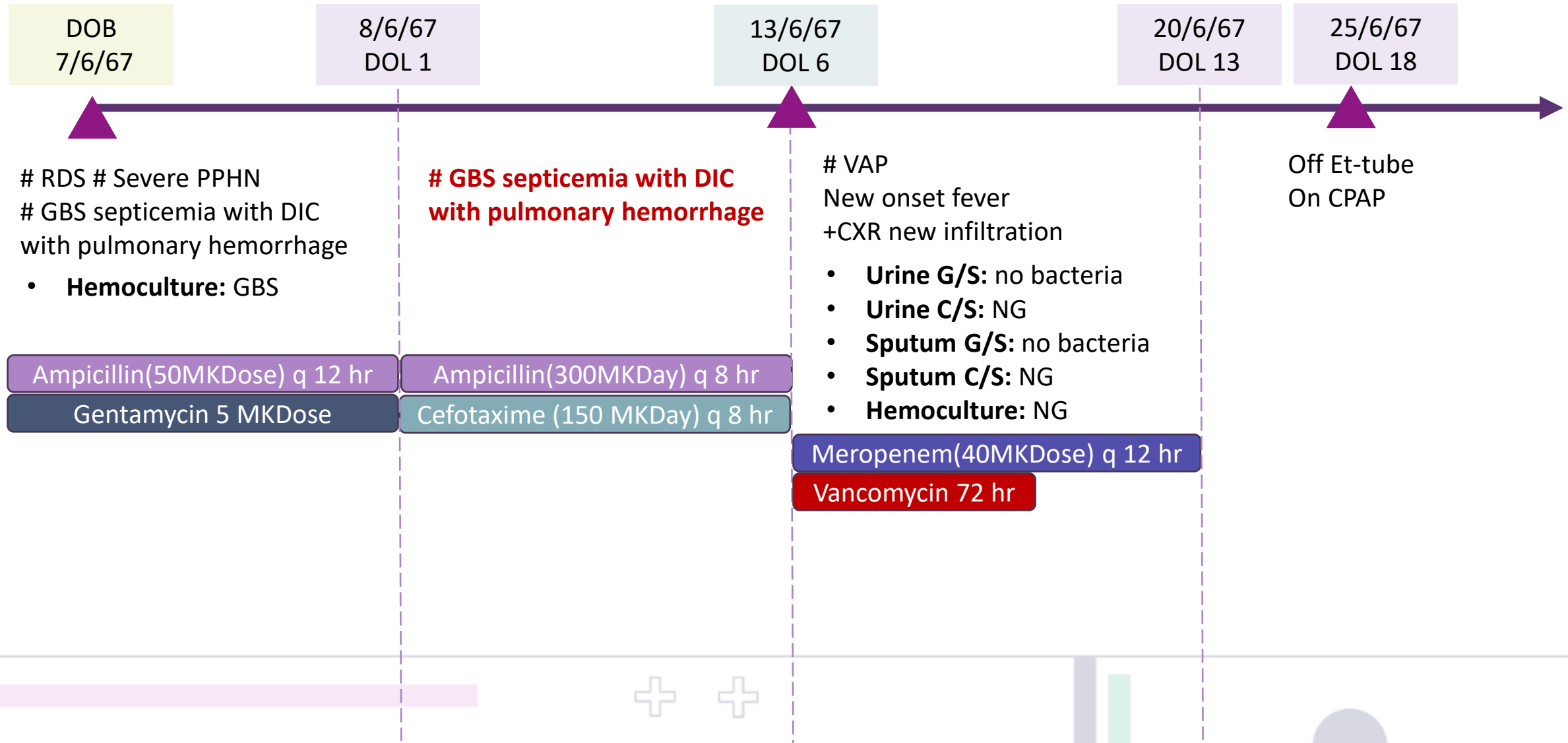
12/6/67



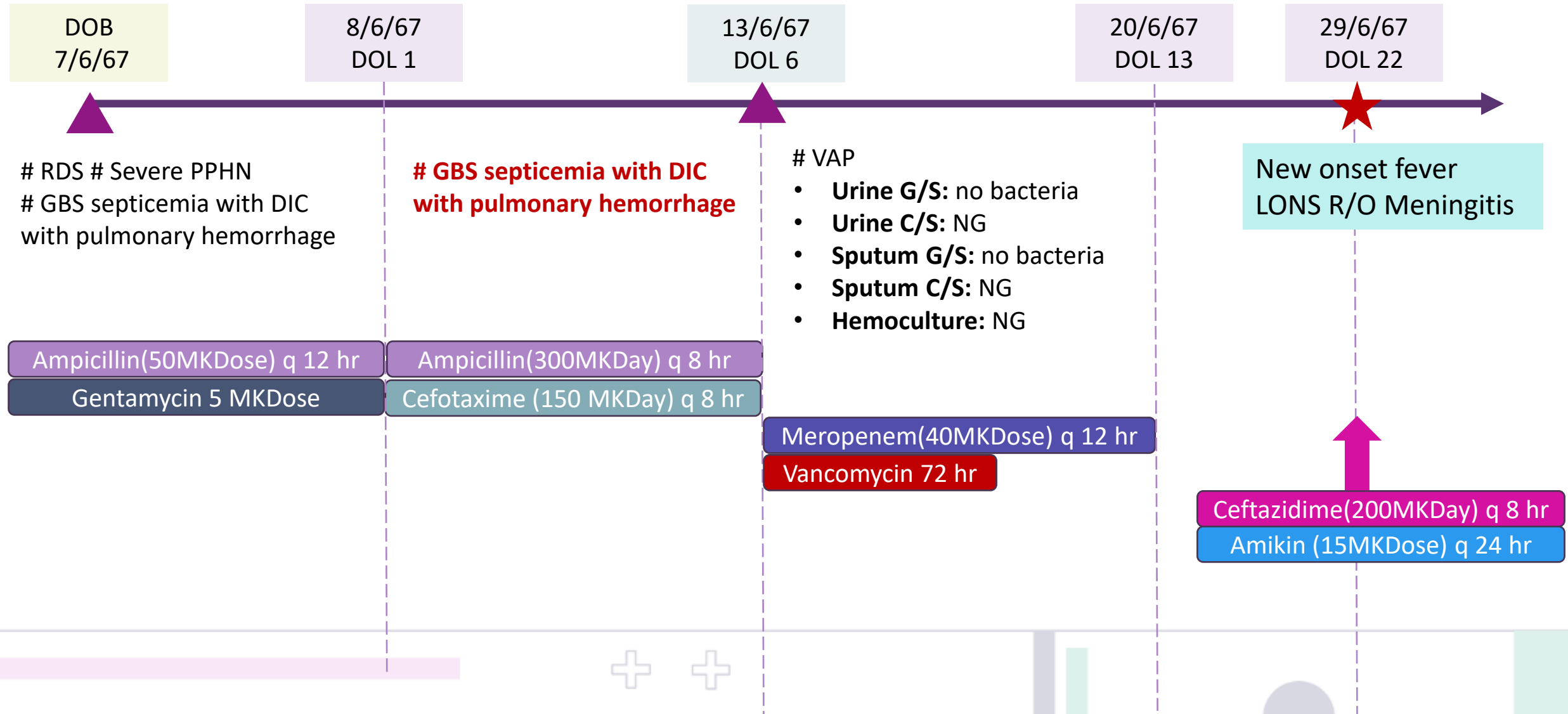
13/6/67



Timeline



Timeline



Lab investigations

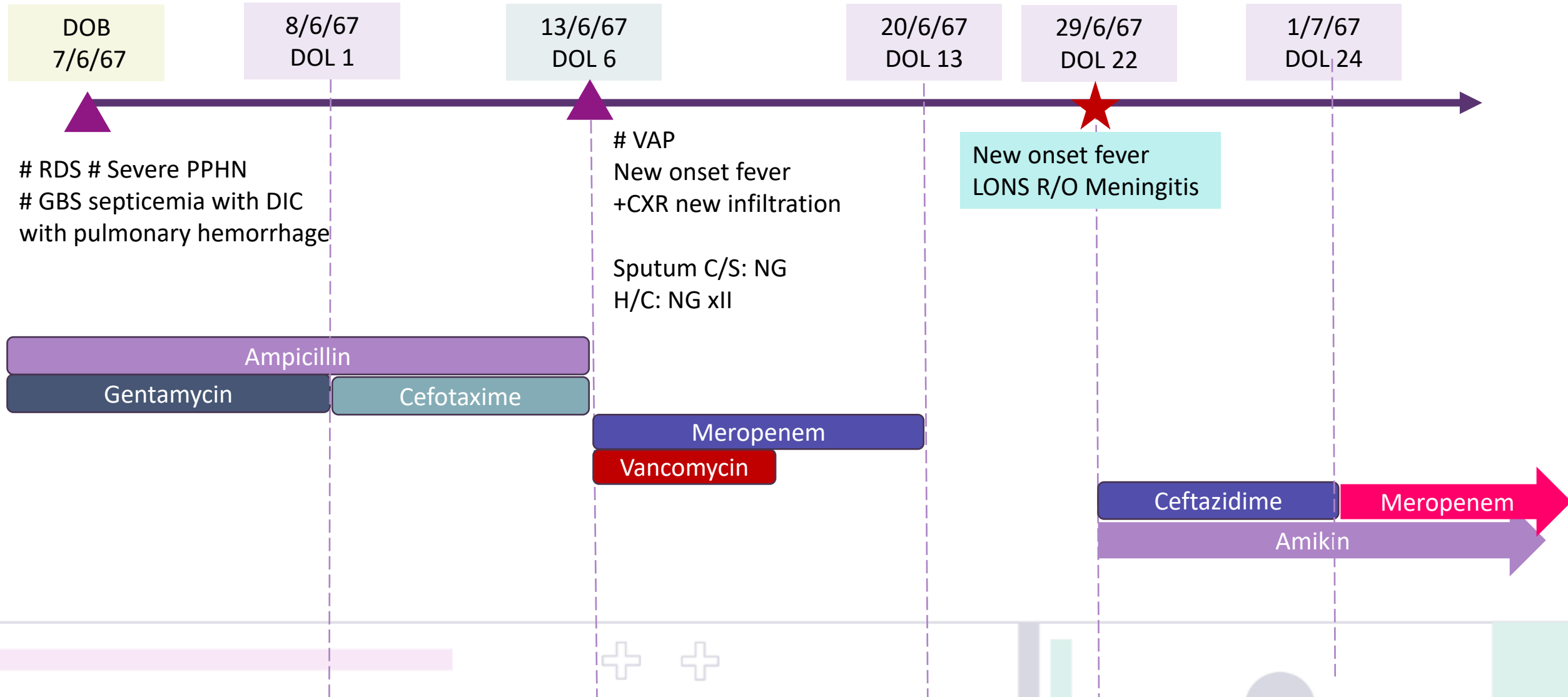
CBC	18/6/67	29/6/67
Hb (g/dl)	10.9	10.3
Hct (%)	32.6	30.4
WBC (cell/cu.mm.)	30,650	16,840
Neutrophil (%)	58.8	51.8
Monocyte (%)	1.7	12.5
Lymphocyte(%)	12.3	27.6
ANC(%)	18,570	8740
Platelet(cell/cu.mm.)	117,000	353,000
MCV	96.2	89.7

Sepsis screen	
Hs-CRP (mg/L)	2.1
Hemoculture	NG x II

Urinalysis	29/6/67
WBC	none
RBC	none
Uspec	1.004
Albumin	negative
Nitrite	negative
Leukocyte esterase	negative
Epithelium	0-1 HPF
Bacteria	Rare HPF



Timeline



Urine Gram Stain and Culture

- **Urine Gram stain:** no bacteria found

#	Qty	Code	Organism	
1	1,000-10,000	ECRCL	Escherichia coli	
#	Code	Antibiotic	MIC	Result
1	AU	Amoxicillin-clavulanic acid	<=2	S
2	TZ	Piperacillin-tazobactam	<=4	S
3	CZ	Ceftazidime	<=0.12	S
4	TX	Ceftriaxone	<=0.25	S
5	FEP	Cefepime	<=0.12	S
6	ETP	Ertapenem	<=0.12	S
7	IP	Imipenem	<=0.25	S
8	ME	Meropenem	<=.25	S
9	AN	Amikacin	2	S
10	G	Gentamicin	<=1	S
11	CI	Ciprofloxacin	<=0.06	S
12	ST	Trimethoprim-sulfamethoxazole	<=1	S

CSF Gram Stain

Gram stain: Few gram negative bacilli



CSF Culture

#	Qty	Code	Organism
1	Many	ECRCL	Escherichia coli
2	Many	KSLPM	Klebsiella pneumoniae
3	Few	SCCGB	Streptococcus agalactiae

#	Code	Antibiotic	MIC	Result
1	AU	Amoxicillin-clavulanic acid	<=2	S
2	TZ	Piperacillin-tazobactam	<=4	S
3	CZ	Ceftazidime	<=0.12	S
4	TX	Ceftriaxone	<=0.25	S
5	FEP	Cefepime	<=0.12	S
6	ME	Meropenem	<=0.25	S
7	AN	Amikacin	2	S
8	G	Gentamicin	<=1	S
9	ST	Trimethoprim-sulfamethoxazole	<=1	S

#	Qty	Code	Organism
1	Many	ECRCL	Escherichia coli
2	Many	KSLPM	Klebsiella pneumoniae
3	Few	SCCGB	Streptococcus agalactiae

#	Code	Antibiotic	MIC	Result
1	AU	Amoxicillin-clavulanic acid	<=2	S
2	TZ	Piperacillin-tazobactam	<=4	S
3	CZ	Ceftazidime	<=0.12	S
4	TX	Ceftriaxone	<=0.25	S
5	FEP	Cefepime	<=0.12	S
6	ME	Meropenem	<=0.25	S
7	AN	Amikacin	<=1	S
8	G	Gentamicin	<=1	S
9	ST	Trimethoprim-sulfamethoxazole	<=1	S

#	Qty	Code	Organism
1	Many	ECRCL	Escherichia coli
2	Many	KSLPM	Klebsiella pneumoniae
3	Few	SCCGB	Streptococcus agalactiae

#	Code	Antibiotic	MIC	Result
1	P	Penicillin	<=0.06	S
2	TX	Ceftriaxone	<=0.12	S
3	VA	Vancomycin	0.5	S



Diagnosis

Preterm female newborn GA 29+2 wk, NL, BBW 1,360 gm

EONS from GBS sepsis with septic shock with DIC

LONS from *Escherichia coli*, *Klebsella pneumoniae*, *Streptococcus agalactiae* meningitis and *Escherichia coli* UTI

Management in this case

- De-escalate to cefotaxime 200 MKDay divided q 8 hrs
 - Plan treat total at least 21 days from first CSF culture negative
- Off Amikin (5/7/67) after repeat UC: no growth (total 7 days)
- Monitoring complication: serial OFC daily

Recommendations for Antimicrobial Therapy

Microorganism	Recommended therapy	Alternative therapies
<i>Streptococcus pneumoniae</i>	Vancomycin plus a third-generation cephalosporin ^{a,b}	Meropenem (C-III), fluoroquinolone ^c (B-II)
<i>Neisseria meningitidis</i>	Third-generation cephalosporin ^a	Penicillin G, ampicillin, chloramphenicol, fluoroquinolone, aztreonam
<i>Listeria monocytogenes</i>	Ampicillin ^d or penicillin G ^d	Trimethoprim-sulfamethoxazole, meropenem (B-III)
<i>Streptococcus agalactiae</i>	Ampicillin ^d or penicillin G ^d	Third-generation cephalosporin ^a (B-III)
<i>Haemophilus influenzae</i>	Third-generation cephalosporin ^a (A-I)	Chloramphenicol, cefepime (A-I), meropenem (A-I), fluoroquinolone
<i>Escherichia coli</i>	Third-generation cephalosporin ^a (A-II)	Cefepime, meropenem, aztreonam, fluoroquinolone, trimethoprim-sulfamethoxazole



CSF C/S: NG

Cefotaxime

Repeat LP

CSF	1/6/67	4/6/67
WBC	53 (34)	9
RBC	19,000	140
PMN (%)	34	33
MN (%)	66	67
Protein (mg/dl)	145	118
Sugar (ratio)	42 (0.42)	54 (0.55)
Gram stain	No bacteria found	No bacteria found
Culture	No growth	No growth
16sRNA	Negative	

Early Onset Neonatal Sepsis

- Occurring in the first 3 days after birth
- Two most common bacterial causes:

GBS, E.coli among term infants

- **E.coli** predominant among preterm

Risk of Early onset GBS disease increased in preterm, PPROM>18 hr, maternal intrapartum fever, intramniotic fluid infection, GBS bacteremia during current pregnancy, previous infant with invasive GBS disease

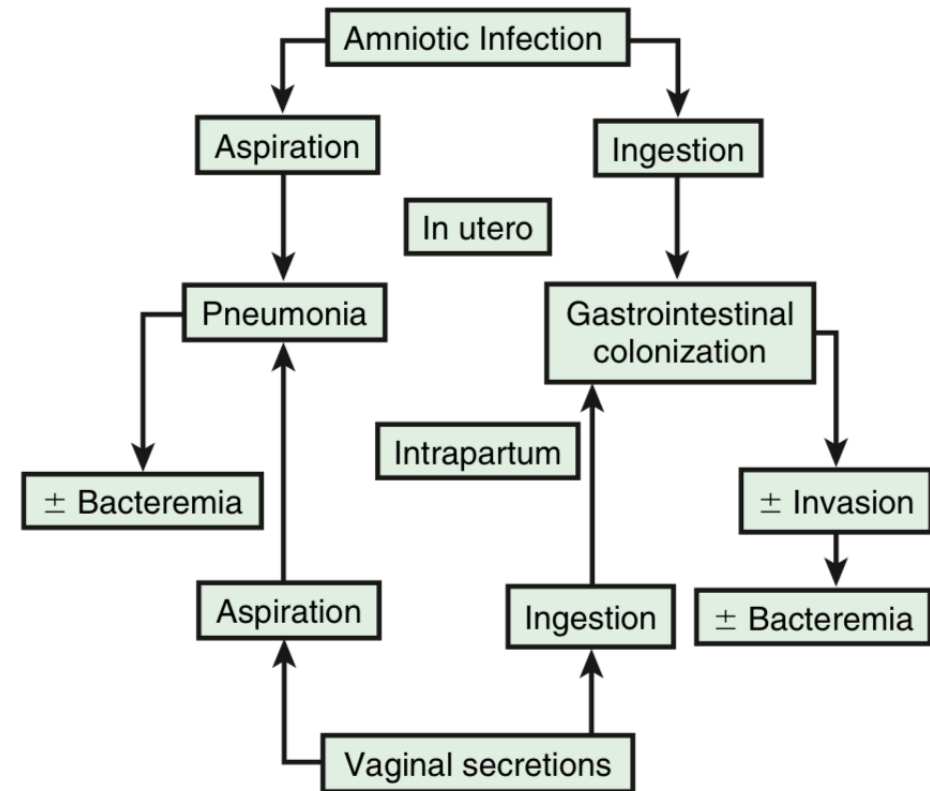
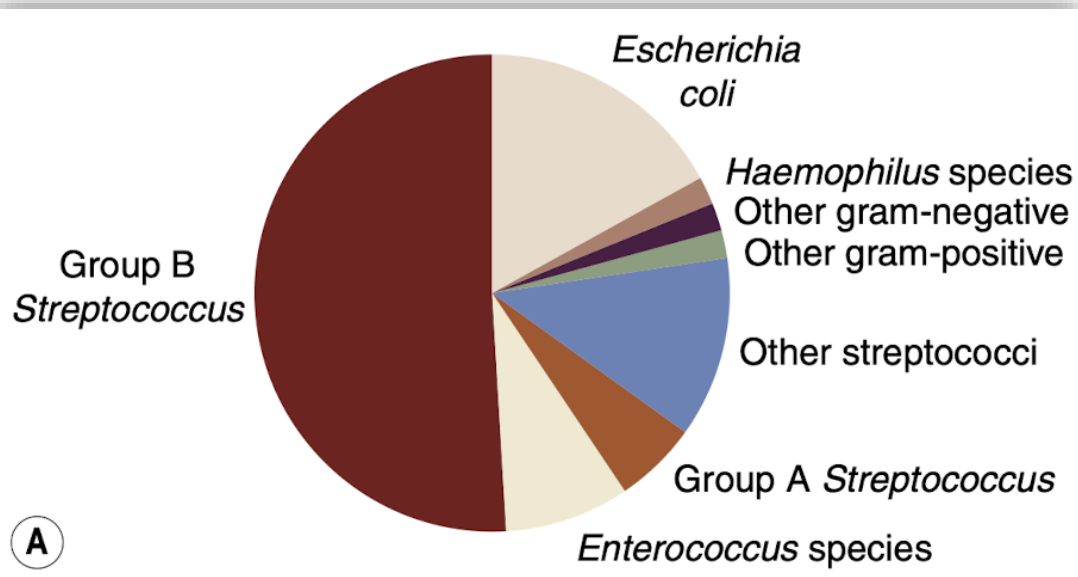


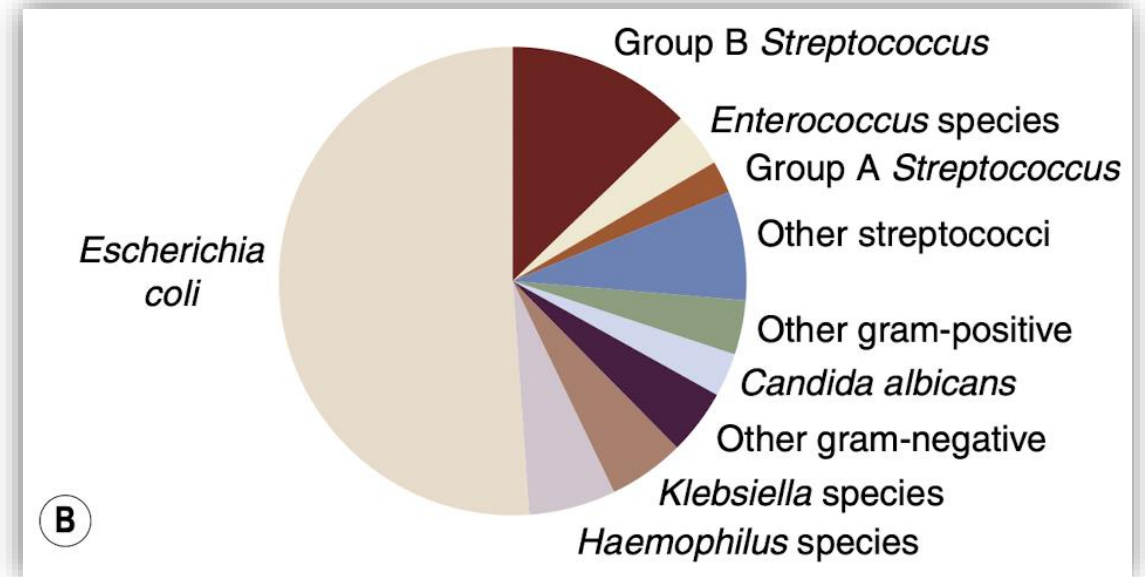
Fig. 148.3 Pathways of ascending or intrapartum infection.

Bacteriology of Organism Isolated from Blood and CSF of Neonate

(High income countries)



Early onset sepsis in term infant (GA ≥ 37 weeks)



Early onset sepsis in preterm infant (GA 22-36 weeks)

Group B Streptococcus Infections

TABLE 119.2 Comparison of Early-, Late-, and Very Late-Onset Group B *Streptococcus* Infections

Feature	Early-Onset	Late-Onset	Very Late-Onset
Age range	<7 days	7–89 days	≥3mo
Median age at onset	1 day	37 days	Unknown
Maternal obstetric complications	Common	Preterm delivery	Preterm delivery
Frequency of gestation <37 wk	Frequent (~20%)	Frequent (~50%)	Almost always
Usual clinical presentations	Septicemia (80%–85%) Meningitis (5%–10%) Pneumonia (10%–15%)	Meningitis (25%–35%) Bloodstream infection without focus (65%) Soft tissue, bone, joint infection, or pneumonia (5%–10%)	Bloodstream infection without focus (common) with a focus (occasional) ^a
Common capsular types	Ia, II, III, V	III (>50%), Ia, V	III, Ia, V
Mortality rate	2%–19% ^b	3%–8%	Low

^aIncludes peritoneum, urinary tract, skin and soft tissue, and central nervous system.

^b<5% in term infants.

Modified from Edwards MS, Nizet V, Baker CJ. Group B streptococcal infections. In: Wilson CB, Nizet V, Maldonado VY, et al. (eds). *Infectious Diseases of the Fetus and Newborn Infant*. 8th ed. Philadelphia, PA: Elsevier Saunders; 2016:411–456.



GBS: Late Onset Infection

- Typically occurs at 3-4 weeks of age
- **Most common presentation: blood stream infection**
 - **Meningitis occur in 30% of cases**
 - Others (rare in era of maternal IAP): osteomyelitis, arthritis, cellulitis-adenitis syndrome
- **Approximately 20% of survivors of neonatal GBS meningitis have moderate to severe neurodevelopmental impairment**



GBS: Recurrent Infection

- Recurrent rate for early-onset disease: 1-2%
- No common risk factors or underlying condition predict recurrence
- Purposed causes: persistent focus, inadequate dose or duration of therapy, reinfection with a second strain or type



Recommendation for empirical antimicrobial therapy

Predisposing factor	Common bacterial pathogens	Antimicrobial therapy
Age		
<1 month	<i>Streptococcus agalactiae</i> , <i>Escherichia coli</i> , <i>Listeria monocytogenes</i> , <i>Klebsiella</i> species	Ampicillin plus cefotaxime or ampicillin plus an aminoglycoside
1–23 months	<i>Streptococcus pneumoniae</i> , <i>Neisseria meningitidis</i> , <i>S. agalactiae</i> , <i>Haemophilus influenzae</i> , <i>E. coli</i>	Vancomycin plus a third-generation cephalosporin ^{a,b}
2–50 years	<i>N. meningitidis</i> , <i>S. pneumoniae</i>	Vancomycin plus a third-generation cephalosporin ^{a,b}
>50 years	<i>S. pneumoniae</i> , <i>N. meningitidis</i> , <i>L. monocytogenes</i> , aerobic gram-negative bacilli	Vancomycin plus ampicillin plus a third-generation cephalosporin ^{a,b}
Head trauma		
Basilar skull fracture	<i>S. pneumoniae</i> , <i>H. influenzae</i> , group A β -hemolytic streptococci	Vancomycin plus a third-generation cephalosporin ^a
Penetrating trauma	<i>Staphylococcus aureus</i> , coagulase-negative staphylococci (especially <i>Staphylococcus epidermidis</i>), aerobic gram-negative bacilli (including <i>Pseudomonas aeruginosa</i>)	Vancomycin plus cefepime, vancomycin plus ceftazidime, or vancomycin plus meropenem
Postneurosurgery	Aerobic gram-negative bacilli (including <i>P. aeruginosa</i>), <i>S. aureus</i> , coagulase-negative staphylococci (especially <i>S. epidermidis</i>)	Vancomycin plus cefepime, vancomycin plus ceftazidime, or vancomycin plus meropenem
CSF shunt	Coagulase-negative staphylococci (especially <i>S. epidermidis</i>), <i>S. aureus</i> , aerobic gram-negative bacilli (including <i>P. aeruginosa</i>), <i>Propionibacterium acnes</i>	Vancomycin plus cefepime, ^c vancomycin plus ceftazidime, ^c or vancomycin plus meropenem ^c



Indication for Repeat LP

- Any patient who has **not responded clinically after 48 hours** of appropriate antimicrobial therapy
- Patient with **pneumococcal meningitis** caused by penicillin- or **cephalosporin-resistant strains**, especially for those who have also received adjunctive dexamethasone therapy
- Neonate with meningitis due to **gram-negative bacilli**
- **Bacterial meningitis in newborn**

Duration of Antimicrobial Therapy

Microorganism	Duration of therapy, days
<i>Neisseria meningitidis</i>	7
<i>Haemophilus influenzae</i>	7
<i>Streptococcus pneumoniae</i>	10–14
<i>Streptococcus agalactiae</i>	14–21
Aerobic gram-negative bacilli ^a	21
<i>Listeria monocytogenes</i>	≥21

★ ^a Duration in the neonate is 2 weeks beyond the first sterile CSF culture or ≥3 weeks, whichever is longer.





Thank You

